Discovery Branch Synthesis

Abducer

KB ref: p-vs-np
Completed branches: 0
Executed branch passes: 25
Proof queue pending: 71
Proof queue completed: 38
Ready for prove: 2
Ready for discovery: 23

Judgment

- All routed discovery branches were executed once. The frontier is now better characterized than expanded: the dominant pressure remains on MCSP carrier realization, formula-size transport, and theorem-activation alignment.
- Next move: Promote only the theorem-shaped branch families into the next prove-or-disprove agenda, and keep the pattern-level branches as discovery guidance rather than treating them as solved proof tasks.

Top Breakthrough Families

- No repeated breakthrough families recorded.

Top Obstruction Classes

- No repeated obstruction classes recorded.

Most Reused MathLib References

- Hardness Magnification Near State-of-the-Art Lower Bounds: 15
- Equivalence and coherent transport theorem for MCSP witness family: 11
- Hardness Magnification for Natural Problems: 11
- A Simple and General Hardness Magnification Theorem: 7
- Polynomially Bounded Proof Systems Characterize NP Equals coNP: 6
- MCSP witness family: major consequences such as NP not in P/poly: 5
- Hitting-Set Generators Yield Black-Box PIT: 5

- Commutative factorization theorem for MCSP obstruction family: 4

Ready For Prove

- disclayer.p-vs-np.family.barrier-aware-transfer: There exists an admissible 'unspecified witness family' such that 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' can be organized into a route preserving 'consequence schema NP-complete-classification' and thereby supports 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'. (priority 2.309; linked queue 3)

- disclayer.p-vs-np.family.explicitness: There exists an admissible 'PIT certificate family' such that 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' can be organized into a route preserving 'PIT semantic object' and thereby supports 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'. (priority 2.107; linked queue 3)

Ready For Discovery

- disclayer.p-vs-np.pattern.invariant-discovery: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. (priority 6.868; linked queue 10)

- disclayer.p-vs-np.pattern.invariant-discovery: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly. (priority 6.868; linked queue 10)

- disclayer.p-vs-np.pattern.regime-localization: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit

lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. (priority 6.758; linked queue 10)

- disclayer.p-vs-np.pattern.regime-localization: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly. (priority 6.758; linked queue 10)

- disclayer.p-vs-np.pattern.hybridization-of-partial-bridges: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. (priority 6.728; linked queue 10)

- disclayer.p-vs-np.pattern.hybridization-of-partial-bridges: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly. (priority 6.728; linked queue 10)

- disclayer.p-vs-np.pattern.counterexample-led-concept-formation: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly. (priority 6.658; linked queue 10)

- disclayer.p-vs-np.pattern.route-decomposition: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly. (priority 6.578; linked queue 10)

- disclayer.p-vs-np.pattern.problem-representation-shift: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly. (priority 6.548; linked queue 10)

- disclayer.p-vs-np.family.realization: If source theorem S holds in regime R, then one can construct an explicit carrier C with admissibility conditions A such that C preserves invariant I and supports a route-specific forcing lemma to target T. (priority 6.443; linked queue 9)
- disclayer.p-vs-np.family.abstraction-search: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'. (priority 6.443; linked queue 9)
- disclayer.p-vs-np.family.abstraction-search: State a route-specific theorem that would resolve the blocked interface and make the next proof obligation materially sharper. (priority 6.443; linked queue 9)
- disclayer.p-vs-np.family.bridge-specificity: Target theorem T is fixed. The real theorem under search is the local activation theorem A that verifies T's hypotheses from current artifacts. (priority 5.669; linked queue 9)
- disclayer.p-vs-np.family.transport: For every admissible carrier C in the nonuniform-circuit regime, there is a transformed carrier C' in the formula-size regime preserving invariant I and retaining enough quantitative strength to imply target T. (priority 5.610; linked queue 8)
- disclayer.p-vs-np.family.barrier-aware-transfer: Either construct a local reduction family G satisfying locality bounds L and preserving witness structure W, or prove that any such G would contradict barrier theorem B. (priority 2.309; linked queue 3)
- disclayer.p-vs-np.family.alignment: Every admissible carrier C from the local route induces an object D in the target theorem's class, and D preserves the quantitative property needed to activate the target theorem. (priority 2.222; linked queue 3)
- disclayer.p-vs-np.family.explicitness: If semantic object C exists for the relevant circuit class, then one can uniformly and efficiently extract an explicit structure G that yields deterministic PIT for that class. (priority 2.107; linked queue 3)
- disclayer.p-vs-np.family.constructive-instantiation: From abstract schema S, construct an explicit family of natural instances E_n satisfying properties P and realizing target theorem T. (priority 1.966; linked queue 3)
- disclayer.p-vs-np.family.semantic-bridge: Local definitional object D has property P if and only if global characterization condition Q holds, with both directions proved at the right level of abstraction. (priority 1.946; linked queue 3)
- disclayer.p-vs-np.family.categorical-reframing: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined

regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute. (priority 0.910; linked queue 12)

- disclayer.p-vs-np.family.categorical-reframing: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute. (priority 0.910; linked queue 12)

- disclayer.p-vs-np.family.equivalence-coherence-reframing: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness. (priority 0.880; linked queue 12)

- disclayer.p-vs-np.family.equivalence-coherence-reframing: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness. (priority 0.880; linked queue 12)

Next Programmes

disclayer.p-vs-np.pattern.invariant-discovery

- Mode: discovery-candidate
- Target: Invariant Discovery Prompt
- Routing priority: 6.868
- Linked queue items: 10
- Headline claim: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.

- Rationale: Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.invariant-discovery

- Mode: discovery-candidate
- Target: Invariant Discovery Prompt
- Routing priority: 6.868
- Linked queue items: 10
- Headline claim: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Rationale: Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.regime-localization

- Mode: discovery-candidate
- Target: Regime Localization Prompt
- Routing priority: 6.758
- Linked queue items: 10
- Headline claim: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Rationale: The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.regime-localization

- Mode: discovery-candidate
- Target: Regime Localization Prompt
- Routing priority: 6.758
- Linked queue items: 10
- Headline claim: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Rationale: The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.hybridization-of-partial-bridges

- Mode: discovery-candidate
- Target: Hybrid Bridge Synthesis Prompt
- Routing priority: 6.728
- Linked queue items: 10
- Headline claim: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Rationale: Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.hybridization-of-partial-bridges

- Mode: discovery-candidate
- Target: Hybrid Bridge Synthesis Prompt
- Routing priority: 6.728

- Linked queue items: 10
- Headline claim: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Rationale: Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.counterexample-led-concept-formation

- Mode: discovery-candidate
- Target: Counterexample-Led Concept Formation Prompt
- Routing priority: 6.658
- Linked queue items: 10
- Headline claim: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Rationale: The frontier is heavily blocked, so near-counterexamples and barrier analogs should be used to induce better concepts rather than only to reject claims.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.route-decomposition

- Mode: discovery-candidate
- Target: Route Decomposition Prompt
- Routing priority: 6.578
- Linked queue items: 10
- Headline claim: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

- Rationale: High-pressure obstruction classes are carrying multiple inferential roles at once, so route decomposition is needed before more local search.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.pattern.problem-representation-shift

- Mode: discovery-candidate
- Target: Problem Representation Shift Prompt
- Routing priority: 6.548
- Linked queue items: 10
- Headline claim: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.
- Rationale: Several blocked routes look misrepresented rather than simply false, so representation change is itself a discovery pattern.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.realization

- Mode: discovery-candidate
- Target: Carrier Realization Prompt
- Routing priority: 6.443
- Linked queue items: 9
- Headline claim: If source theorem S holds in regime R, then one can construct an explicit carrier C with admissibility conditions A such that C preserves invariant I and supports a route-specific forcing lemma to target T.
- Rationale: Turn a typed but schematic carrier into an explicit mathematical object family with admissibility conditions and a forcing theorem.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.abstraction-search

- Mode: discovery-candidate
- Target: Abstraction Search Prompt
- Routing priority: 6.443
- Linked queue items: 9

- Headline claim: There exists an admissible 'MCSP witness family' such that 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' can be organized into a route preserving 'MCSP complexity profile in nonuniform-circuits' and thereby supports 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.'.
- Rationale: Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.abstraction-search

- Mode: discovery-candidate
- Target: Abstraction Search Prompt
- Routing priority: 6.443
- Linked queue items: 9
- Headline claim: State a route-specific theorem that would resolve the blocked interface and make the next proof obligation materially sharper.
- Rationale: Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.bridge-specificity

- Mode: discovery-candidate
- Target: Activation Theorem Prompt
- Routing priority: 5.669
- Linked queue items: 9
- Headline claim: Target theorem T is fixed. The real theorem under search is the local activation theorem A that verifies T's hypotheses from current artifacts.
- Rationale: Stop treating the target theorem as the open problem and relocate search to the bridge that would activate it.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.transport

- Mode: discovery-candidate
- Target: Transport Principle Prompt
- Routing priority: 5.610
- Linked queue items: 8
- Headline claim: For every admissible carrier C in the nonuniform-circuit regime, there is a transformed carrier C' in the formula-size regime preserving invariant I and retaining enough quantitative strength to imply target T.
- Rationale: Find a theorem that preserves the relevant invariant across a cross-model move into formula-size.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.barrier-aware-transfer

- Mode: prove-candidate
- Target: Barrier-Aware Transfer Prompt
- Routing priority: 2.309
- Linked queue items: 3
- Headline claim: There exists an admissible 'unspecified witness family' such that 'There exist many problems L in combinatorial optimisation/graph theory such that L is NP-complete under polynomial-time many-one reductions.' can be organized into a route preserving 'consequence schema NP-complete-classification' and thereby supports 'Under"local"polynomial-time reductions (where each output bit is computable with very small time), MCSP is not NP-complete. More broadly, the paper develops evidence and conditional consequences suggesting NP-hardness of MCSP under various reductions would entail unexpected class separations.'.
- Rationale: Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.barrier-aware-transfer

- Mode: discovery-candidate
- Target: Barrier-Aware Transfer Prompt
- Routing priority: 2.309
- Linked queue items: 3

- Headline claim: Either construct a local reduction family G satisfying locality bounds L and preserving witness structure W, or prove that any such G would contradict barrier theorem B.
- Rationale: Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.alignment

- Mode: discovery-candidate
- Target: Alignment Theorem Prompt
- Routing priority: 2.222
- Linked queue items: 3
- Headline claim: Every admissible carrier C from the local route induces an object D in the target theorem's class, and D preserves the quantitative property needed to activate the target theorem.
- Rationale: Identify or prove an equivalence between the current carrier class and the object class required by the target theorem.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.explicitness

- Mode: prove-candidate
- Target: Explicitness Upgrade Prompt
- Routing priority: 2.107
- Linked queue items: 3
- Headline claim: There exists an admissible 'PIT certificate family' such that 'A set Hsubseteq mathbb F ^n is a hitting set for a class mathcal C of n -variate circuits if for every nonzero polynomial p computed by some circuit in mathcal C , there exists mathbf a in H with p(mathbf a)neq 0.' can be organized into a route preserving 'PIT semantic object' and thereby supports 'If mathrm PIT in Pclass (or even suitable infinitely-often derandomisations in nondeterministic subexponential time, in standard variants), then at least one of the following holds: NEXP not in P/poly quad or quad Permanent does not have polynomial-size arithmetic circuits.'.
- Rationale: Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.explicitness

- Mode: discovery-candidate
- Target: Explicitness Upgrade Prompt
- Routing priority: 2.107
- Linked queue items: 3
- Headline claim: If semantic object C exists for the relevant circuit class, then one can uniformly and efficiently extract an explicit structure G that yields deterministic PIT for that class.
- Rationale: Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.constructive-instantiation

- Mode: discovery-candidate
- Target: Constructive Instantiation Prompt
- Routing priority: 1.966
- Linked queue items: 3
- Headline claim: From abstract schema S, construct an explicit family of natural instances E_n satisfying properties P and realizing target theorem T.
- Rationale: Turn an abstract schema into explicit natural example families or prove that the schema cannot instantiate naturally in the current way.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.semantic-bridge

- Mode: discovery-candidate
- Target: Semantic Bridge Prompt
- Routing priority: 1.946
- Linked queue items: 3
- Headline claim: Local definitional object D has property P if and only if global characterization condition Q holds, with both directions proved at the right level of abstraction.
- Rationale: Bridge a local definitional object to a global characterization theorem.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.categorical-reframing

- Mode: discovery-candidate
- Target: Categorical Reframing Prompt
- Routing priority: 0.910
- Linked queue items: 12
- Headline claim: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Rationale: Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.categorical-reframing

- Mode: discovery-candidate
- Target: Categorical Reframing Prompt
- Routing priority: 0.910
- Linked queue items: 12
- Headline claim: There exists a mediating object built from 'MCSP witness family' such that the route from 'A hardness magnification theorem has the form:"A barely-nontrivial lower bound for an explicit problem Q in a weak model implies a major lower bound/separation in a strong model."Typically, Q is a meta-problem (e.g. a Gap-MCSP variant) and"major"means consequences such as NP not in P/poly.' to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.' factors through typed morphisms preserving 'MCSP complexity profile in nonuniform-circuits' and making the relevant square commute.
- Rationale: Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.equivalence-coherence-reframing

- Mode: discovery-candidate
- Target: Equivalence / Coherence Reframing Prompt
- Routing priority: 0.880
- Linked queue items: 12
- Headline claim: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.
- Rationale: Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`

disclayer.p-vs-np.family.equivalence-coherence-reframing

- Mode: discovery-candidate
- Target: Equivalence / Coherence Reframing Prompt
- Routing priority: 0.880
- Linked queue items: 12
- Headline claim: There exists an equivalence witness identifying 'MCSP witness family' with the target-side object class relevant to 'In refined regimes, lower bounds for Gap-MCSP slightly beyond known frontiers (e.g. N^ 1+varepsilon -type circuit lower bounds for suitable promise gaps and input length N=2^n) imply major consequences such as NP not in P/poly.', and 'MCSP complexity profile in nonuniform-circuits' transports coherently along that witness.
- Rationale: Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.
- Report: `C:\{}files\{}projects\{}abducer\{}abducer\{}artifacts\{}p-vs-np\{}discovery-branch-`